

B.Tech/M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Second & Third Semester

21CSC101T - OBJECT ORIENTED DESIGN AND PROGRAMMING

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

- i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours**Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

PART - A (20 × 1 = 20 Marks)		Marks	BL	CO
Answer all Questions				
1.	Which of the following is a valid declaration of a static variable in C++? (A) static int x; (B) int static x; (C) int x; (D) stat int x;	1	1	1
2.	Which access specifier allows members to be accessed within the same class and derived classes? (A) public (B) private (C) protected (D) internal	1	1	1
3.	In C++, which keyword is used to declare a constant variable? (A) constant (B) const (C) final (D) readonly	1	1	1
4.	What does UML stand for? (A) Unified Modeling Language (B) Universal Modeling Language (C) Unified Method Language (D) Unified Management Language	1	1	1
5.	In operator overloading, which keyword is used to define the operator function? (A) op (B) operator (C) overload (D) oper	1	1	2
6.	C++ supports _____ inheritance, which allows a derived class to inherit from many base classes, even if the base classes are unrelated. (A) Multilevel (B) Single level (C) Multiple (D) Hierarchical	1	1	2
7.	_____ emphasizes on time sequence of messages from one object to another. (A) Sequence diagram (B) Collaboration diagram (C) Class diagram (D) State diagram	1	1	2
8.	Which among the following best describes constructor overloading? (A) Defining one constructor in each class of program (B) Defining more than one constructor in single class (C) Defining more than one constructor in single class with different signature (D) Defining destructor with each constructor	1	1	2
9.	A function which has no implementation is called (A) Pure virtual function (B) virtual function (C) friend function (D) any user defined function	1	1	3
10.	Elements of round corner boxes to represent the situation of objects (A) Lifeline (B) Iteration (C) transition (D) state	1	1	3

11.	Collaboration and sequence diagram are considered as	1	1	3
	(A) interaction diagram (B) non-structural			
	(C) structural diagram (D) non-behavioural			
12.	When a virtual function is redefined by the derived class, it is called	1	1	3
	(A) Overloading (B) Overriding			
	(C) Rewriting (D) Overwriting			
13.	Which feature of C++ allows for writing generic functions and classes?	1	1	4
	(A) Inheritance (B) Function Overloading			
	(C) Templates (D) Pointers			
14.	_____ is used to handle exceptional situations during program execution.	1	1	4
	(A) break and continue (B) try and catch			
	(C) if-else statements (D) switch-case			
15.	What is the purpose of the 'finally' block in exception handling?	1	1	4
	(A) To catch exceptions (B) To throw exceptions			
	(C) To clean up resources (D) To define custom exceptions			
16.	_____ UML diagram illustrates the system's internal structure, showing how software components are connected or related.	1	1	4
	(A) Use Case Diagram (B) Component Diagram			
	(C) Deployment Diagram (D) State Diagram			
17.	Which container provides constant time insertion and deletion at both the beginning and end?	1	1	5
	(A) Vector (B) List			
	(C) Dequeue (D) Array			
18.	_____ header file must be included to use the transform() algorithm in C++.	1	1	5
	(A) <algorithm> (B) <iostream>			
	(C) <vector> (D) <iterator>			
19.	_____ container stores elements in a sorted order based on keys.	1	1	5
	(A) Vector (B) List			
	(C) Stack (D) Map			
20.	_____ is the purpose of an iterator in C++ STL.	1	1	5
	(A) To store elements in a container (B) To iterate over the elements of a container			
	(C) To sort elements in a container (D) To search for elements in a container			

PART - B (5 × 8 = 40 Marks)

Answer **all** Questions

21.	(a) Illustrate class and object with example.	8	2	1
	(OR)			
	(b) Develop a use case diagram to show the bank customer relationship and explain the notations.			
22.	(a) Write a program to add 2 complex number using friend function and + operator.	8	3	2
	(OR)			
	(b) Draw UML sequence diagram for Library Management diagram for borrowing a book.			

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|-----|--|---|---|---|
| 23. | (a) What are the types of inheritance? Explain with neat diagram and example program. | 8 | 3 | 3 |
| | (OR) | | | |
| | (b) Draw the activity diagram with swim lanes for railway reservation system. The passenger should be able to book ticket in online railway reservation. The diagram should include fork, joint, decision etc. | | | |
| 24. | (a) Write a CPP program to find the largest of two numbers using function templates. | 8 | 3 | 4 |
| | (OR) | | | |
| | (b) Discuss Component and Deployment diagram. Illustrate the notations of Component and Deployment diagram for Vehicle Registration System. | | | |
| 25. | (a) Elaborate in detail about Associative Container: Map, Multi-map and Set, Muti-set with example program. | 8 | 3 | 5 |
| | (OR) | | | |
| | (b) Explain in detail about the components that make up the Standard Template Libraries (STL). | | | |

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Questions

Marks BL CO

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|-----|---|----|---|---|
| 26. | A Book shop maintains details of books that are being sold like bookid, title, author, price. Whenever a customer gives the above details display them to the user and if they provide a discount % along with the other details mentioned above, calculate the new discounted price, and display all the details. Implement the above scenario using constructor overloading. | 15 | 3 | 2 |
| 27. | Create a class to print the area of a square and a rectangle. The class has two functions with the same name but a different number of parameters. The function for printing the area of a rectangle has two parameters which are its length and breadth respectively while the other function for printing the area of the square has one parameter which is the side of the square. | 15 | 3 | 4 |

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